A Bounded-Domain Collapse for Asymptotically Holder Nonexpansive Type Maps

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Source And Target

The source paper is C. S. Barroso and C. S. R. da Silva, *Fixed point results for asymptotically Holder nonexpansive type mappings*, arXiv:2605.30678. After proving a Goebel–Kirk style theorem for asymptotically Holder nonexpansive type mappings under the hypothesis $\epsilon_0(X) < 1$, the paper says that the extent to which other classical results, including those of Dominguez Benavides and Lorenzo Ramirez, extend to this framework remains a subject for future investigation.

The supporting source is T. Dominguez Benavides and P. Lorenzo Ramirez, *Fixed points for mappings of asymptotically nonexpansive type*, arXiv:2201.02802, Fixed Point Theory 24 (2023), 569–582. In particular, their Theorem 5.3 states that if X has uniform normal structure, C is a bounded convex subset of X, and $T : C \to C$ is asymptotically nonexpansive type, then there exists $x \in C$ such that $T^n x \to x$.

Definitions

Let C be a bounded subset of a Banach space X. A map $T : C \to C$ is of asymptotically Holder nonexpansive type if there is a sequence $\alpha_n \in (0, 1]$ with $\alpha_n \to 1$ such that, for every $x \in C$,

$$\limsup_{n\to\infty} \sup_{y\in C} (\|T^n x - T^n y\| - \|x - y\|^{\alpha_n}) \leq 0.$$

The ordinary asymptotically nonexpansive type condition is the same assertion with $\|x - y\|$ in place of $\|x - y\|^{\alpha_n}$.

Proof Intuition

The Holder version looks weaker at small scales, because $t^\alpha \geq t$ on $[0, 1]$ when $0 < \alpha \leq 1$. But on any bounded interval the functions t^α converge uniformly to t as $\alpha \to 1$. Therefore the Holder-type inequality differs from the usual asymptotically nonexpansive type inequality by an additive error that tends to zero uniformly over the whole domain. Once this reduction is made, the fixed-point machinery already known for asymptotically nonexpansive type maps applies verbatim.

Lemma 1 (Bounded-domain collapse)**.** *Let C be a bounded subset of a Banach space X, and let $\alpha_n \in (0, 1]$ satisfy $\alpha_n \to 1$. If $T : C \to C$ is of asymptotically Holder nonexpansive type with exponent sequence (α_n), then T is of ordinary asymptotically nonexpansive type.*

Proof. Put $D = \operatorname{diam}(C) < \infty$. For $n \geq 1$, define

$$\eta_n = \sup_{0 \leq t \leq D} \big(t^{\alpha_n} - t\big).$$

The sequence (α_n) is eventually contained in some compact interval $[\alpha_0, 1] \subset (0, 1]$. The map $(t, \alpha) \mapsto t^\alpha$ is continuous on $[0, D] \times [\alpha_0, 1]$, hence uniformly continuous there; therefore $\eta_n \to 0$.

Fix $x \in C$. For every $y \in C$ and every n,

$$\|T^n x - T^n y\| - \|x - y\| \leq (\|T^n x - T^n y\| - \|x - y\|^{\alpha_n}) + \eta_n.$$

Taking the supremum over $y \in C$ and then the limsup over n gives

$$\limsup_{n\to\infty} \sup_{y \in C} (\|T^n x - T^n y\| - \|x - y\|) \leq 0 + 0 = 0.$$

This is the ordinary asymptotically nonexpansive type condition. □

Theorem 1 (Uniform-normal-structure transfer)**.** *Let X be a Banach space with uniform normal structure, let $C \subset X$ be a bounded convex subset, and let $T : C \to C$ be of asymptotically Holder nonexpansive type. Then there exists $x \in C$ such that $T^n x \to x$. Consequently, under any continuity hypothesis that turns such a convergent orbit into a fixed point, for instance if some fixed iterate T^N is continuous, T has a fixed point.*

Proof. By Lemma 1, T is asymptotically nonexpansive type in the ordinary sense. Dominguez Benavides–Lorenzo Ramirez Theorem 5.3 therefore applies and yields $x \in C$ with $T^n x \to x$.

If T^N is continuous, then $T^{n+N} x = T^N(T^n x) \to T^N x$, while also $T^{n+N} x \to x$. Hence $T^N x = x$. The orbit of x is then periodic with period dividing N, but the full orbit $T^n x$ converges to x; therefore all points in the finite cycle equal x, in particular $Tx = x$. □

Corollary 1. *Every bounded-domain fixed-point or orbit-convergence theorem whose hypothesis is the ordinary asymptotically nonexpansive type condition immediately applies to asymptotically Holder nonexpansive type mappings with $\alpha_n \to 1$. In particular, the 2026 theorem under $\epsilon_0(X) < 1$ is part of a larger uniform-normal-structure transfer.*

Limitations And Novelty Check

This packet does not settle the Dowling–Lennard–Turett renorming question for spaces containing ℓ_1; the source paper explicitly separates that general asymptotically nonexpansive problem from its Holder-class result. The result here addresses the bounded-domain Holder-type framework and the source paper's broader invitation to extend classical asymptotic-type fixed point theorems.

The cheap run indexes were searched for arXiv:2605.30678 and for combinations of "Holder", "asymptotically nonexpansive type", and "uniform normal structure". No duplicate packet was found. A local full-source search found the supporting 2022/2023 theorem but no packet recording this bounded-domain collapse for the 2026 Holder-type paper.

References

[1] C. S. Barroso and C. S. R. da Silva, *Fixed point results for asymptotically Holder nonexpansive type mappings*, arXiv:2605.30678, 2026.

[2] T. Dominguez Benavides and P. Lorenzo Ramirez, *Fixed points for mappings of asymptotically nonexpansive type*, Fixed Point Theory 24 (2023), 569–582; arXiv:2201.02802.